

PHY604 Lecture 10

September 25, 2025

Today's lecture: Linear Algebra

- Gaussian elimination
- LU decomposition
- Iterative methods

By hand: Forward elimination

- 1. **Eliminate x_1 from second and third equation.** Add first equation to the second and subtract twice the first equation from the third:

$$x_1 + x_2 + x_3 = 6$$

$$3x_2 + x_3 = 9$$

$$-2x_2 - x_3 = -7$$

- 2. **Eliminate x_2 from third equation.** Multiply the second equation by $(-2/3)$ and subtract it from the third

$$x_1 + x_2 + x_3 = 6$$

$$3x_2 + x_3 = 9$$

$$-\frac{1}{3}x_3 = -1$$

By hand: Back substitution

$$x_1 + x_2 + x_3 = 6$$

$$3x_2 + x_3 = 9$$

$$-\frac{1}{3}x_3 = -1$$

- 3. Solve for $x_3 = 3$.
- 4. Substitute x_3 into the second equation to get $x_2 = 2$
- 5. Substitute x_3 and x_2 into the first equation to get $x_1 = 1$
- In general, for N variables and N equations:
 - Use forward elimination make the last equation provide the solution for x_N
 - Back substitute from the N th equation to the first
 - Scales like N^3 (can do better for “sparse” equations)

Pitfalls of Gaussian substitution: Roundoff errors

- Consider a different example (also from Garcia):

$$\epsilon x_1 + x_2 + x_3 = 5$$

$$x_1 + x_2 = 3$$

$$x_1 + \quad + x_3 = 4$$

- First, let's take $\epsilon \rightarrow 0$ and solve:

Subtract second from third:

$$x_2 + x_3 = 5$$

$$x_1 + x_2 = 3$$

$$-x_2 + x_3 = 1$$

Add first to third:

$$x_2 + x_3 = 5$$

$$x_1 + x_2 = 3$$

$$2x_3 = 6$$

Back substitute:

$$x_2 = 2$$

$$x_1 = 1$$

$$x_3 = 3$$

Roundoff error example: Now solve with ε

- Forward elimination starts by multiplying first equation by $1/\varepsilon$ and subtracting it from second and third:

$$\varepsilon x_1 + x_2 + x_3 = 5$$

$$(1 - 1/\varepsilon)x_2 - (1/\varepsilon)x_3 = 3 - 5/\varepsilon$$

$$- (1/\varepsilon)x_2 + (1 - 1/\varepsilon)x_3 = 4 - 5/\varepsilon$$

- Clearly have an issue if ε is near zero, e.g., if $C - 1/\varepsilon \rightarrow -1/\varepsilon$ for C order unity:

$$\varepsilon x_1 + x_2 + x_3 = 5$$

$$- (1/\varepsilon)x_2 - (1/\varepsilon)x_3 = -5/\varepsilon$$

$$- (1/\varepsilon)x_2 - (1/\varepsilon)x_3 = -5/\varepsilon$$

Cannot solve,
now have two
equations, three
unknowns

Simple fix: Pivoting

- Interchange the order of the equations before performing the forward elimination

$$x_1 + x_2 = 3$$

$$\epsilon x_1 + x_2 + x_3 = 5$$

$$x_1 + \quad + x_3 = 4$$

- Now the first step of forward elimination gives us:

$$x_1 + x_2 = 3$$

$$(1 - \epsilon)x_1 + x_3 = 5 - 3\epsilon$$

$$-x_2 + x_3 = 1$$

- Now we round off:

$$x_1 + x_2 = 3$$

$$x_1 + x_3 = 5$$

$$-x_2 + x_3 = 1$$

Same as when we initially took ϵ to 0.

Gaussian elimination with pivoting

- Partial-pivoting:
 - Interchange of rows to move the one with the largest element in the current column to the top
 - (Full pivoting would allow for row and column swaps—more complicated)
- Scaled pivoting
 - Consider largest element relative to all entries in its row
 - Further reduces roundoff when elements vary in magnitude greatly
- Row echelon form: This is the form that the matrix is in after forward elimination

Matrix determinants with Gaussian elimination

- Once we have done forward substitution and obtained a row echelon matrix it is trivial to calculate the determinant:

$$\det(\mathbf{A}) = (-1)^{N_{\text{pivot}}} \prod_{i=1}^N A_{ii}^{\text{row-echelon}}$$

- Every time we pivoted in the forward substitution, we change the sign

Matrix inverse with Gaussian elimination

- We can also use Gaussian elimination to find the inverse of a matrix
- We would like to find $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$
- We can use Gaussian elimination to solve: $\mathbf{A} \mathbf{x}_i = \mathbf{e}_i$
 - $\mathbf{e}_i$ is a column of the identity:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \end{bmatrix}, \dots, \quad \mathbf{e}_N = \begin{bmatrix} \vdots \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

- $\mathbf{x}_i$ is a column of the inverse:

$$\mathbf{A}^{-1} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \mathbf{x}_3 \quad \dots \quad \mathbf{x}_N]$$

Singular matrix

- If a matrix has a vanishing determinant, then the system is not solvable
- Common way for this to enter, one equation in the system is a linear combination of some others
- Not always easy to detect from the start

Singular and close to singular matrices

- Condition number: Measures how close to singular we are
 - How much $\mathbf{x}$ would change with a small change in $\mathbf{b}$

$$\text{cond}(\mathbf{A}) = \|\mathbf{A}\| \|\mathbf{A}^{-1}\|$$

- Requires defining a norm of $\mathbf{A}$
 - https://en.wikipedia.org/wiki/Matrix_norm
- See, e.g., numpy implementation:
 - <https://numpy.org/doc/stable/reference/generated/numpy.linalg.cond.html>

- Rule of thumb: $\frac{\|\mathbf{x}^{\text{exact}} - \mathbf{x}^{\text{calc}}\|}{\|\mathbf{x}^{\text{exact}}\|} \simeq \text{cond}(\mathbf{A}) \cdot \epsilon^{\text{machine}}$

Tridiagonal and banded matrices

- We saw this type of matrix when solving for cubic spline coefficients:

$$\begin{pmatrix} 4\Delta x & \Delta x & & & \\ \Delta x & 4\Delta x & \Delta x & & \\ & \Delta x & 4\Delta x & \Delta x & \\ & & \ddots & \ddots & \ddots \\ & & & \Delta x & 4\Delta x & \Delta x \\ & & & & \Delta x & 4\Delta x \end{pmatrix} \begin{pmatrix} p_1'' \\ p_2'' \\ p_3'' \\ \vdots \\ p_{n-2}'' \\ p_{n-1}'' \end{pmatrix} = \frac{6}{\Delta x} \begin{pmatrix} f_0 - 2f_1 + f_2 \\ f_1 - 2f_2 + f_3 \\ f_2 - 2f_3 + f_4 \\ \vdots \\ f_{n-3} - 2f_{n-2} + f_{n-1} \\ f_{n-2} - 2f_{n-1} + f_n \end{pmatrix}$$

- Often come up in physical situations
- These types of matrices can be efficiently solved with Gaussian elimination

Gaussian elimination for banded matrices

- Only need to do Gaussian elimination steps for m nonzero elements below given row (m is less than the number of diagonal bands)
- Example:

$$\begin{pmatrix} 2 & 1 & 0 & 0 \\ 3 & 4 & -5 & 0 \\ 0 & -4 & 3 & 5 \\ 0 & 0 & 1 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2.5 & -5 & 0 \\ 0 & -4 & 3 & 5 \\ 0 & 0 & 1 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2.5 & -5 & 0 \\ 0 & 0 & -5 & 5 \\ 0 & 0 & 1 & 3 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2.5 & -5 & 0 \\ 0 & 0 & -5 & 5 \\ 0 & 0 & 0 & 4 \end{pmatrix}$$

Today's lecture: Linear Algebra

- Gaussian elimination
- LU decomposition
- Iterative methods

LU decomposition (Newman Ch. 6)

- Often happens that we would like to solve: $\mathbf{A}\mathbf{x}_i = \mathbf{v}_i$ for the same $\mathbf{A}$ but many $\mathbf{v}$
 - For example, our implementation for the inverse
 - Wasteful to do Gaussian elimination over and over, we will always get the same row echelon matrix, just $\mathbf{v}_i$ will be different
 - Instead, we should keep track of operations we did to $\mathbf{v}_1$ and use them over and over

- Consider a general 4 x 4 matrix:

$$\begin{pmatrix} a_{00} & a_{01} & a_{02} & a_{03} \\ a_{10} & a_{11} & a_{12} & a_{13} \\ a_{20} & a_{21} & a_{22} & a_{23} \\ a_{30} & a_{31} & a_{32} & a_{33} \end{pmatrix}$$

- Let's perform Gaussian elimination

LU decomposition: First GE step

- Write the first step of the GE as:

$$\frac{1}{a_{00}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ -a_{10} & a_{00} & 0 & 0 \\ -a_{20} & 0 & a_{00} & 0 \\ -a_{30} & 0 & 0 & a_{00} \end{pmatrix} \begin{pmatrix} a_{00} & a_{01} & a_{02} & a_{03} \\ a_{10} & a_{11} & a_{12} & a_{13} \\ a_{20} & a_{21} & a_{22} & a_{23} \\ a_{30} & a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} 1 & b_{01} & b_{02} & b_{03} \\ 0 & b_{11} & b_{12} & b_{13} \\ 0 & b_{21} & b_{22} & b_{23} \\ 0 & b_{31} & b_{32} & b_{33} \end{pmatrix}$$

- Where the b 's are some linear combination of a coefficients
- The first matrix on the LHS is a lower triangular matrix we call:

$$\mathbf{L}_0 \equiv \frac{1}{a_{00}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ -a_{10} & a_{00} & 0 & 0 \\ -a_{20} & 0 & a_{00} & 0 \\ -a_{30} & 0 & 0 & a_{00} \end{pmatrix}$$

LU decomposition: Second LU step

$$\frac{1}{b_{11}} \begin{pmatrix} b_{11} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -b_{21} & b_{11} & 0 \\ 0 & -b_{31} & 0 & b_{11} \end{pmatrix} \begin{pmatrix} 1 & b_{01} & b_{02} & b_{03} \\ 0 & b_{11} & b_{12} & b_{13} \\ 0 & b_{21} & b_{22} & b_{23} \\ 0 & b_{31} & b_{32} & b_{33} \end{pmatrix} = \begin{pmatrix} 1 & c_{01} & c_{02} & c_{03} \\ 0 & 1 & c_{12} & c_{13} \\ 0 & 0 & c_{22} & c_{23} \\ 0 & 0 & c_{32} & c_{33} \end{pmatrix}$$

$$\mathbf{L}_1 \equiv \frac{1}{b_{11}} \begin{pmatrix} b_{11} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -b_{21} & b_{11} & 0 \\ 0 & -b_{31} & 0 & b_{11} \end{pmatrix}$$

LU decomposition: Last two steps for 4x4 matrix

$$\mathbf{L}_2 \equiv \frac{1}{c_{22}} \begin{pmatrix} c_{22} & 0 & 0 & 0 \\ 0 & c_{22} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -c_{32} & c_{22} \end{pmatrix}, \quad \mathbf{L}_3 \equiv \frac{1}{d_{33}} \begin{pmatrix} d_{33} & 0 & 0 & 0 \\ 0 & d_{33} & 0 & 0 \\ 0 & 0 & d_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- So, we can write:

$$\mathbf{L}_3 \mathbf{L}_2 \mathbf{L}_1 \mathbf{L}_0 \mathbf{A} = \mathbf{L}_3 \mathbf{L}_2 \mathbf{L}_1 \mathbf{L}_0 \mathbf{v}$$

- Afterwards, the equation is ready for back substitution
- Mathematically identical to Gaussian elimination, but we only have to find $\mathbf{L}_0$ - $\mathbf{L}_3$ once, and then we can operate on many $\mathbf{v}$'s

Slightly different formulation of LU decomposition

- From the properties of upper triangular matrices (same holds for lower):
 - Product of two upper triangular matrices is an upper triangular matrix.
 - Inverse of an upper triangular matrix is an upper triangular matrix
- Consider the lower-diagonal matrix **L** and the upper-diagonal matrix **U**:

$$\mathbf{L} = \mathbf{L}_0^{-1} \mathbf{L}_1^{-1} \mathbf{L}_2^{-1} \mathbf{L}_3^{-1}, \quad \mathbf{U} = \mathbf{L}_3 \mathbf{L}_2 \mathbf{L}_1 \mathbf{L}_0 \mathbf{A}$$

- Then trivially: **LU = A**, so for **Ax = v**, we can write **LUx = v**

Expression for $\mathbf{L}$

- We can confirm that for our 4 x 4 example,

$$\mathbf{L}_0^{-1} = \begin{pmatrix} a_{00} & 0 & 0 & 0 \\ a_{10} & 1 & 0 & 0 \\ a_{20} & 0 & 1 & 0 \\ a_{30} & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{L}_1^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & b_{11} & 0 & 0 \\ 0 & b_{21} & 1 & 0 \\ 0 & b_{31} & 0 & 1 \end{pmatrix}, \quad \mathbf{L}_2^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & c_{22} & 0 \\ 0 & 0 & c_{32} & 1 \end{pmatrix}, \quad \mathbf{L}_3^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & d_{33} \end{pmatrix}$$

- Multiplying together we get

$$\mathbf{L} = \begin{pmatrix} a_{00} & 0 & 0 & 0 \\ a_{10} & b_{11} & 0 & 0 \\ a_{20} & b_{21} & c_{22} & 0 \\ a_{30} & b_{31} & c_{32} & d_{33} \end{pmatrix}$$

Solving the equation with **L** and **U**

- Break into two steps:
 - 1. **Ly = v** can be solved by back substitution:

$$\begin{pmatrix} l_{00} & 0 & 0 & 0 \\ l_{10} & l_{11} & 0 & 0 \\ l_{20} & l_{21} & l_{22} & 0 \\ l_{30} & l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} v_0 \\ v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

- 2. Now solve **Ux = y** by back substitution:

$$\begin{pmatrix} u_{00} & u_{01} & u_{02} & u_{03} \\ 0 & u_{11} & u_{12} & u_{13} \\ 0 & 0 & u_{22} & u_{23} \\ 0 & 0 & 0 & u_{33} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

Some comments about LU decomposition

- Most common method for solving simultaneous equations
- Decomposition needs to be done once, then only back substitution is needed for different $\mathbf{v}$
- In general, still may need to pivot
 - Every time you swap rows, you have to do the same to $\mathbf{L}$
 - Need to perform the same sequence of swaps on $\mathbf{v}$

Today's lecture:

More on Linear Algebra

- Gaussian elimination
- LU decomposition
- Iterative methods

Jacobi and Gauss-Seidel iterative methods

- Gaussian elimination is a **direct** method
- We can also use an **iterative** method
 - Choose an initial guess and converge to better and better guesses
 - E.g., Jacobi or Gauss Seidel, Newton methods
 - Can be much more efficient for very large systems
 - Often puts restrictions on the form of the matrix for guaranteed convergence

Jacobi iterative method

- Starting with a linear system:
$$\begin{array}{cccc} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & = & b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & = & b_2 \\ \vdots & & \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n & = & b_n \end{array}$$

- Pick initial guesses $\mathbf{x}^k$, solve equation i for i th unknown to get an improved guess:

$$x_1^{k+1} = -\frac{1}{a_{11}}(a_{12}x_2^k + a_{13}x_3^k + \cdots + a_{1n}x_n^k - b_1)$$

$$x_2^{k+1} = -\frac{1}{a_{22}}(a_{21}x_1^k + a_{23}x_3^k + \cdots + a_{2n}x_n^k - b_2)$$

$$\vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots$$

$$x_n^{k+1} = -\frac{1}{a_{nn}}(a_{n1}x_1^k + a_{n2}x_2^k + \cdots + a_{n,n-1}x_{n-1}^k - b_n)$$

Jacobi iterative method

- We can write an element-wise formula for $\mathbf{x}$:

$$x_i^{k+1} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j^k \right)$$

- Or:

$$\mathbf{x}^{k+1} = \mathbf{D}^{-1} (\mathbf{b} - (\mathbf{A} - \mathbf{D})\mathbf{x}^k)$$

- Where $\mathbf{D}$ is a diagonal matrix constructed from the diagonal elements of $\mathbf{A}$
- Convergence is guaranteed if matrix is diagonally dominant (but works in other cases):

$$a_{ii} > \sum_{j=1, j \neq i}^N |a_{ij}|$$

Multivariate Newton's method

- We can generalize Newton's method for equations with several variables
 - Can be used when we no longer have a linear system
 - Cast the problem as one of root finding
- Consider the vector function: $\mathbf{f}(\mathbf{x}) = [f_1(\mathbf{x}) \quad f_2(\mathbf{x}) \quad \dots \quad f_N(\mathbf{x})]$
- Where the unknowns are: $\mathbf{x} = [x_1 \quad x_2 \quad \dots \quad x_N]$
- Revised guess from initial guess $\mathbf{x}^{(0)}$: $\mathbf{x}_1 = \mathbf{x}_0 - \mathbf{f}(\mathbf{x}_0)\mathbf{J}^{-1}(\mathbf{x}_0)$
 - $\mathbf{J}^{-1}$ is the inverse of the Jacobian matrix:

$$J_{ij}(\mathbf{x}) = \frac{\partial f_i(\mathbf{x})}{\partial x_j}$$

- To avoid taking the inverse at each step, solve with Gaussian substitution:

$$\mathbf{J}\delta\mathbf{x}^k = -\mathbf{f}(\mathbf{x}^k)$$

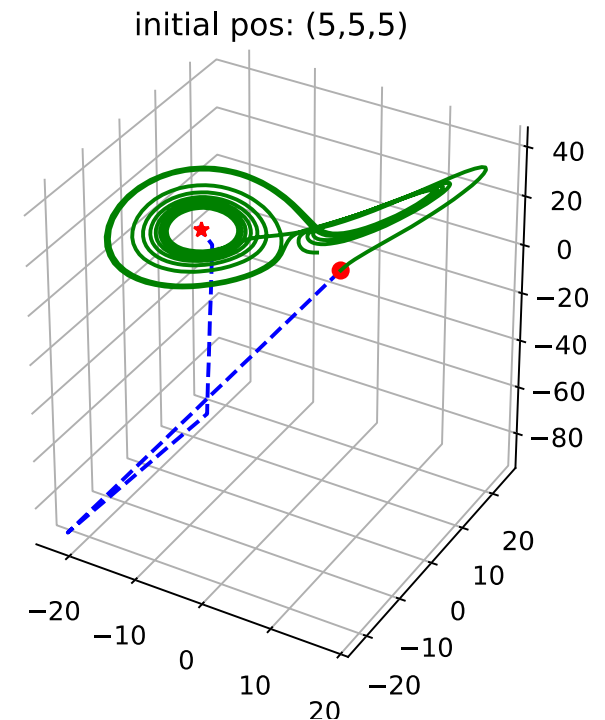
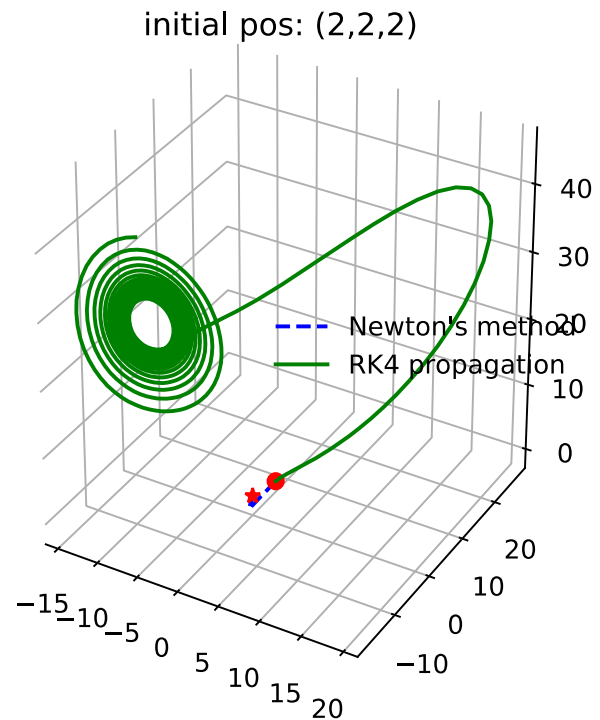
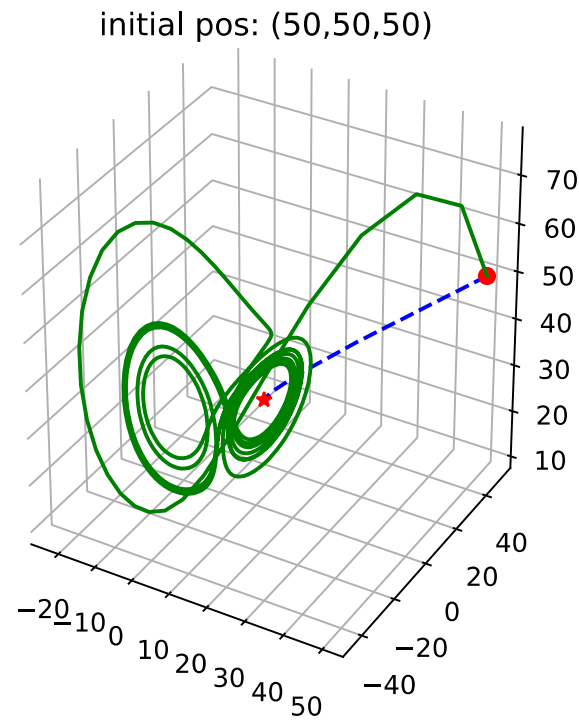
Example: Lorenz model (Garcia Sec. 4.3)

- Lorenz system:
$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x) \\ \frac{dy}{dt} &= rx - y - xz \\ \frac{dz}{dt} &= xy - bz\end{aligned}$$

- σ , r , and b are positive constants
- If we want steady-state, we can propagate with, e.g., 4th order RK
- Steady-state directly given by roots of Lorenz system:

$$\mathbf{f}(x, y, z) = \begin{pmatrix} \sigma(y - x) \\ rx - y - xz \\ xy - bz \end{pmatrix} = 0 \qquad \mathbf{J} = \begin{pmatrix} -\sigma & \sigma & 0 \\ r - z & -1 & -x \\ y & x & -b \end{pmatrix}$$

Lorenz model steady-state: Newton versus 4th order RK



Finding the extrema of multivariable functions

- To get extrema of $g(\mathbf{x})$, Must solve the nonlinear equation:

$$\mathbf{f}(\mathbf{x}) = \nabla g(\mathbf{x}) = 0$$

- Need to ensure that $g(\mathbf{x})$ continually decreases if we want the minima, or continually increases if we want the maximum, modify the Jacobian in Newton's method

$$J_{ij}(\mathbf{x}) = \frac{\partial f_i(\mathbf{x})}{\partial x_j} + \mu \delta_{ij}$$

- μ is small and positive to make sure $\mathbf{A}$ is positive definite:
- Popular scheme involves updating μ with each step: $\mathbf{w}^T \mathbf{A} \mathbf{w} \geq 0 \quad \forall \mathbf{w} \neq 0$

$$\mathbf{A}_k = \mathbf{A}_{k-1} + \frac{\mathbf{y}\mathbf{y}^T}{\mathbf{y}^T \mathbf{w}} - \frac{\mathbf{A}_{k-1} \mathbf{w} \mathbf{w}^T \mathbf{A}_{k-1}}{\mathbf{w}^T \mathbf{A}_{k-1} \mathbf{w}}, \quad \mathbf{w} = \mathbf{x}_k - \mathbf{x}_{k-1}, \quad \mathbf{y} = \mathbf{f}_k - \mathbf{f}_{k-1}$$

- BFGS method (Broyden, Fletcher, Goldfarb, Shanno)

After class tasks

- Homework 2 due on Wednesday Oct. 1
- Readings:
 - Newman Ch. 6
 - Garcia Ch. 4
 - Pang Sec. 5.3